



*From Algorithm to the Clinic:*  
**A Preference-Aware CDS for Shared  
Decision Making**

Team HAC

# Roadmap (1/1)



# We would like to thank the authors of the papers referenced in this study.

1. **Athey, S., & Wager, S. (2021).** *Efficient policy learning*. arXiv. <https://doi.org/10.48550/arXiv.1702.02896>
2. **Bach, K., et al. (2019).** Design of a clinician dashboard to facilitate co-decision making in the management of non-specific low back pain. *Journal of Intelligent Information Systems*, 53(1), 167–189. <https://doi.org/10.1007/s10844-018-0540-6>
3. **Broekhuizen, H., IJzerman, M. J., Hauber, A. B., & Groothuis-Oudshoorn, C. G. M. (2017).** Weighing clinical evidence using patient preferences: An application of probabilistic multi-criteria decision analysis. *PharmacoEconomics*, 35(2), 259–269. <https://doi.org/10.1007/s40273-016-0467-z>
4. **Butler, J. L. (2016).** *Using patient preferences to estimate optimal treatment strategies for competing outcomes* (Doctoral dissertation, University of North Carolina at Chapel Hill). Carolina Digital Repository. <https://doi.org/10.17615/8p58-pf19>
5. **Dakin, H., et al. (2018).** Mapping to estimate health-state utility values: A systematic review of development and reporting practices. *Medical Decision Making*, 38(7), 894–905. <https://doi.org/10.1177/0272989X18792737>
6. **Fusiak, J., Sarpari, K., Ma, I., Mansmann, U., & Hoffmann, V. S. (2025).** Practical applications of methods to incorporate patient preferences into medical decision models: A scoping review. *BMC Medical Informatics and Decision Making*, 25, 109. <https://doi.org/10.1186/s12911-025-02945-5>
7. **Lenert, L., et al. (2014).** A model to support shared decision making in electronic health records systems. *Medical Decision Making*, 34(8), 987–995. <https://doi.org/10.1177/0272989X14550102>
8. **Pöyhönen, M., & Hämäläinen, R. P. (1997).** *On the convergence of multiattribute weighting methods* (Systems Analysis Laboratory Research Report). Helsinki University of Technology.
9. **Romero-Brufau, S., et al. (2020).** Implementation of artificial intelligence–based clinical decision support to reduce hospital readmissions at a regional hospital. *JMIR Medical Informatics*, 8(7), e16308. <https://doi.org/10.2196/16308>
10. **Weyant, C., & Brandeau, M. L. (2021).** Personalization of medical treatment decisions: Simplifying complex models while maintaining patient health outcomes. *Medical Decision Making*, 41(7), 783–796. <https://doi.org/10.1177/0272989X211037921>



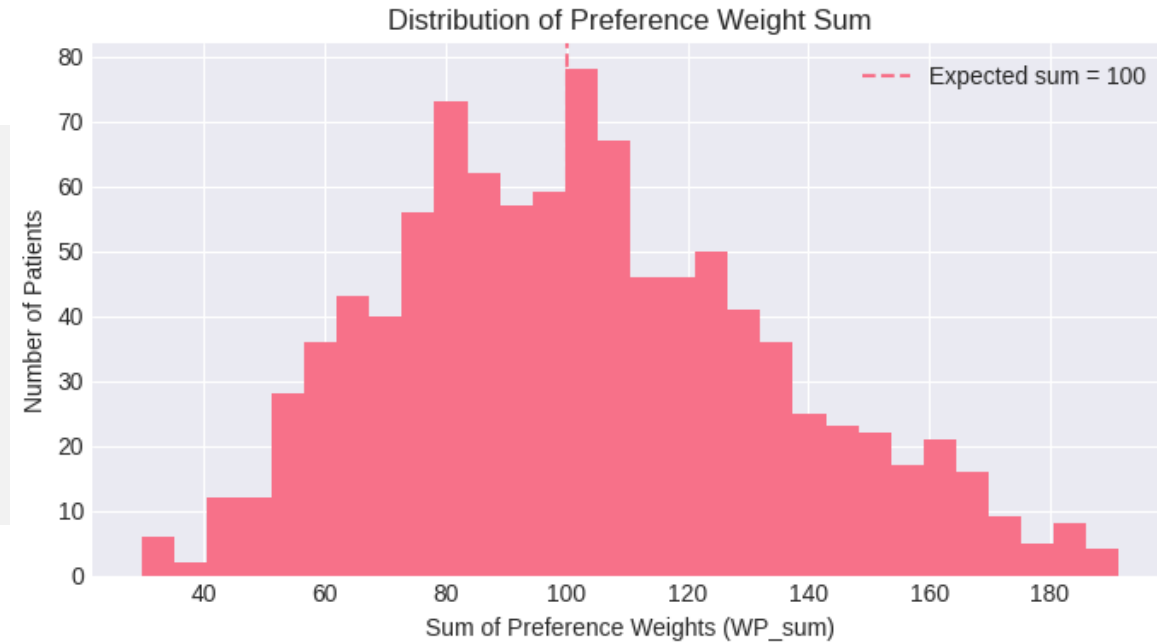
# Research Question *(1/1)*

How can we transparently combine predicted outcomes and patient preferences to support shared decision making in clinical practice?

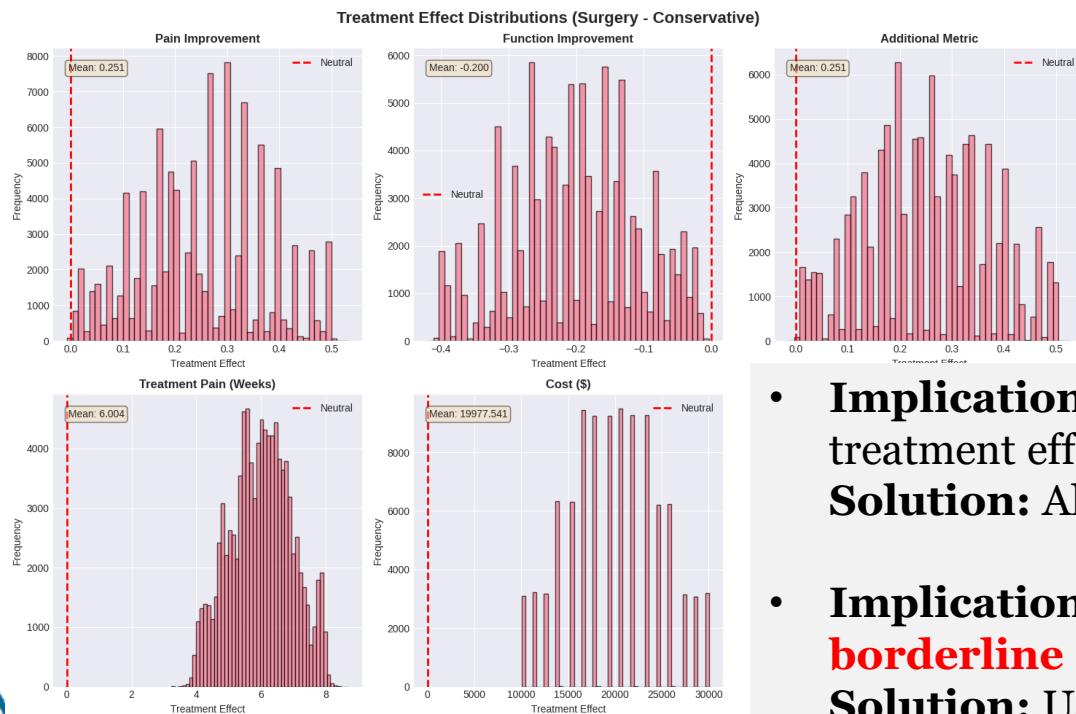


# Data Exploratory (1/3)

- **Implication:** Patients did not consistently allocate exactly 100 “chips”
- **Issue:** Bias results, as patients with higher totals would disproportionately influence the decision score
- **Solution:** Preference weights were normalized to sum to 100

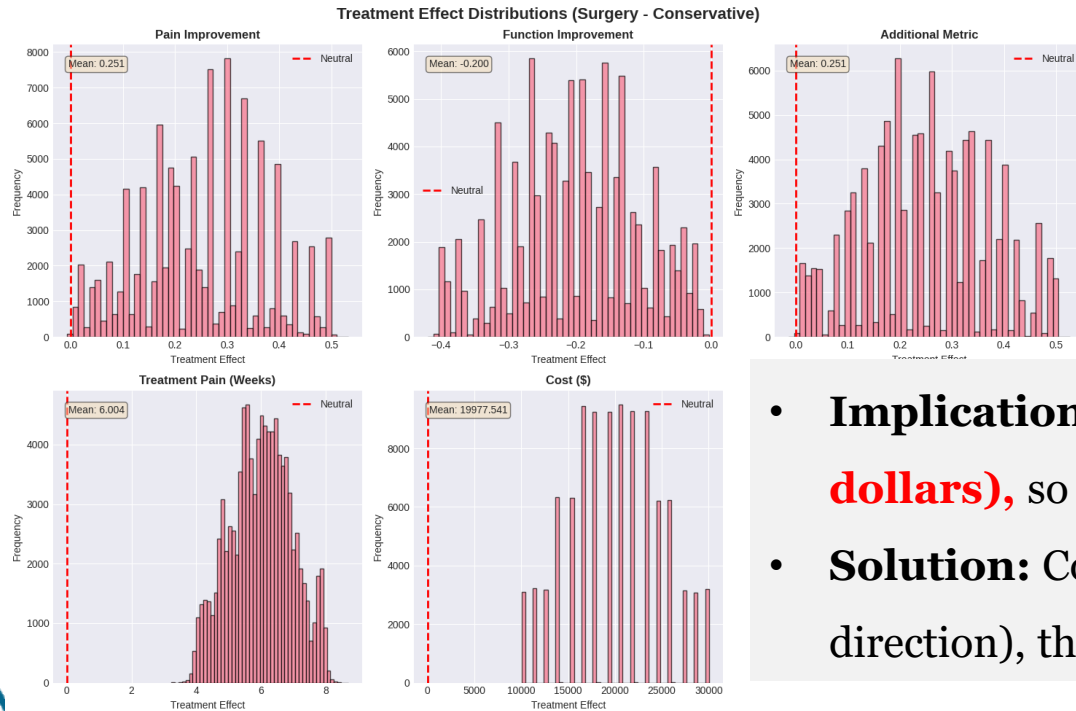


# Implication (2/3)



- **Implication: Outcomes use different definition and scales**, so raw treatment effects are not comparable.  
**Solution:** Align directions and normalize before aggregation.
- **Implication: Heterogeneity and preference sensitivity create many borderline cases.**  
**Solution:** Use individualized effects, a decision margin, and flag cases for SDM.

# Implication (3/3)



- **Implication:** The five outcomes use different units (scores, weeks, dollars), so raw treatment effects can't be compared or added fairly.
- **Solution:** Convert all effects to a common dollar-value scale (after aligning benefit direction), then combine them using normalized patient preference weights.



# Decision framework (1/2)

## 1 Preference-Weighted Utility

**Key Idea:** Decisions should reflect both predicted outcomes and patient values.

*Weyant & Brandeau (2021), Medical Decision Making*

### Framework Implementation:

→ Combine predicted treatment effects with **patient-specific preference weights** to compute utility differences ( $utility = outcomes \times preferences$ ).

## 2 Outcome Normalization

**Key Idea:** Outcomes on different units cannot be combined directly.

*Drummond et al. (2015), Methods for the Economic Evaluation of Health Care Programmes*

### Framework Implementation:

→ **Align outcome directions** and apply **scaling/normalization** before aggregation.

## 3 Patient-Specific Treatment Effects

**Key Idea:** Population averages may mislead individual treatment decisions.

*Kent et al. (2018), Annals of Internal Medicine*

### Framework Implementation:

→ Base recommendations on **individualized treatment effects (CATEs)** rather than average effects.

## 4 Borderline Decisions & SDM

**Key Idea:** Decision support should surface uncertainty, not hide it.

*Elwyn et al. (2012), BMJ*

### Framework Implementation:

→ Apply a **decision margin** and explicitly flag **borderline cases** to support shared decision making.



# Decision framework (2/2)

## Estimate Treatment Effects

Patient-specific outcomes for Surgery vs Conservative

## Apply Patient Preferences

Weight outcomes using WP1–WP5

## Compute Utility Difference

$\Delta U = \text{Surgery} - \text{Conservative}$

## Decision Rule

**Recommend | Borderline | Trade-offs | Defer to SDM**

## Sensitivity Check

**Flag preference-sensitive cases**

# Key Methodology (1/4)

## OutcomeScaler (WTP-Based Outcome Scaling)

### Why:

*Converting to \$ value makes trade-offs comparable and easy to interpret.*

### How:

#### Set WTP anchors (from health system research):

- Long-term pain: \$25,000
- Treatment-related pain: \$15,000
- Function: \$25,000

#### Convert TE → \$ value per outcome

- TE1 →  $\text{pain\_improvement\_value} = \text{TE1} \times \$25\text{k}$
- TE2 →  $\text{function\_improvement\_value} = \text{TE2} \times \$25\text{k}$
- TE3 →  $\text{additional\_quality\_value} = \text{TE3} \times \text{avg}(\$25\text{k}, \$25\text{k})$
- TE4 (weeks) →  $\text{treatment\_pain\_cost} = -(\text{TE4}/52 \text{ weeks}) \times \$15\text{k}$
- TE5 (\$) →  $\text{direct\_cost} = -\text{TE5}$

# Key Methodology (2/4)

## Preference-Weighted Utility Integration

### Why:

*We need a consistent way to turn patient preferences + outcome values into one transparent decision score.*

### How:

#### Input per patient

- Treatment effects row TE1–TE5
- Preference weights WP1–WP5\_normalized

#### Scale outcomes

- Returns dollar-value outcomes (positive = surgery better)

#### Apply preference weights

- Convert each weight from % to decimal:  $\text{weight} = \text{WP\_normalized} / 100$
- Multiply:  $\text{contribution} = \text{weight} \times \text{outcome\_value}$

#### Sum contributions → total utility

- $\text{total\_utility} = \sum \text{contribution\_k}$
- Also store a breakdown of each component (weight, value, contribution) for explainability

# Key Methodology (3/4)

## Utility-Based Treatment Recommendation

### Why:

*Translate utility scores into clear, actionable recommendations with confidence.*

### How:

- If  $\text{total\_utility} > 0 \rightarrow \text{Surgery}$ ; else  $\rightarrow \text{Conservative}$
- Use utility magnitude to flag borderline cases
  - Compute  $\text{abs\_utility} = |\text{total\_utility}|$
  - If  $< \$2,000 \rightarrow \text{Low (Borderline)}$   
If  $< \$4,000 \rightarrow \text{Moderate}$   
Else  $\rightarrow \text{High}$
- Surface key trade-offs and cost considerations for shared decision making

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CLINICAL DECISION SUPPORT RECOMMENDATION

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Patient ID: 1.0

RECOMMENDED TREATMENT: Surgery  
Confidence Level: Low - Borderline  
Utility Difference: \$124.68

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KEY TRADE-OFFS

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1. Surgery shows \$4,327 advantage in pain improvement value
2. Conservative care shows \$5,158 advantage in function improvement value
3. Surgery shows \$8,348 advantage in additional quality value
4. Conservative care shows \$1,975 advantage in treatment pain cost
5. Conservative care shows \$19,335 advantage in direct cost

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CLINICAL QUALIFICATIONS FOR SDM

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1. ⚠️ **BORDERLINE CASE:** Both treatments show similar overall benefit. Strong emphasis on patient values and shared decision-making.
2. Mixed outcomes: Surgery is better in some areas but worse in others. Discuss specific trade-offs with patient.
3. Cost is a major factor in this recommendation. Consider patient's financial situation and insurance coverage.

# Key Methodology (4/4)

## Preference Sensitivity Analysis

### Why:

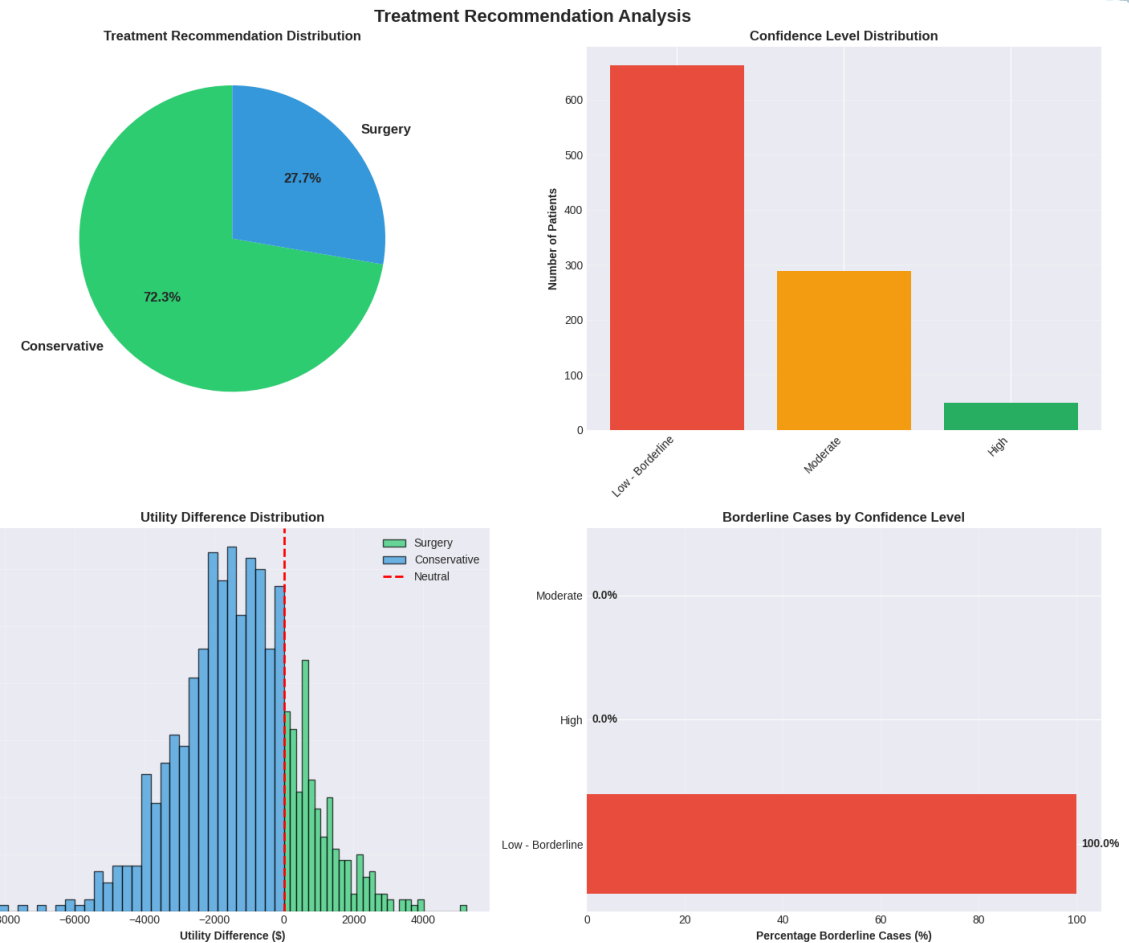
*Check whether the recommendation is **stable** or changes when patient preferences shift.*

### How:

- Compute the base recommendation and utility score
- For each WPk: increase it by +20%, reduce others proportionally, then renormalize to 100
- Recompute recommendation
- Mark as stable if no variation changes the treatment decision

# Predicted Outcome (1/1)

- **Conservative care dominates:** ~72% of patients are recommended non-surgical treatment.
- **Many borderline cases:** Most decisions lie near the threshold, indicating the need for shared decision making.
- **Clear cases are few:** High-confidence recommendations are rare but decisive when present.

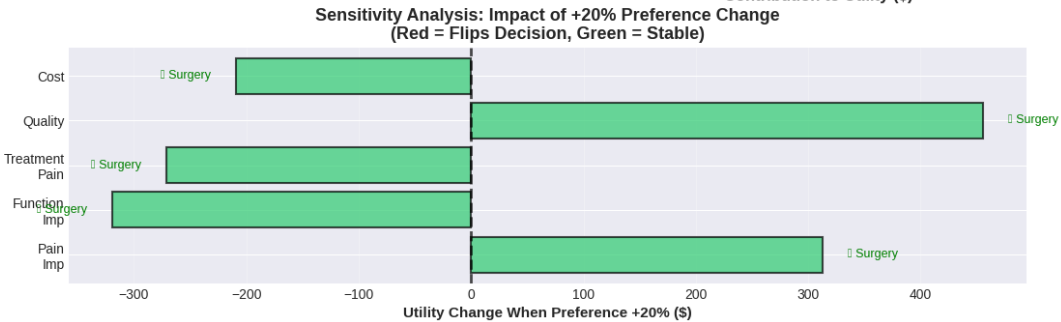
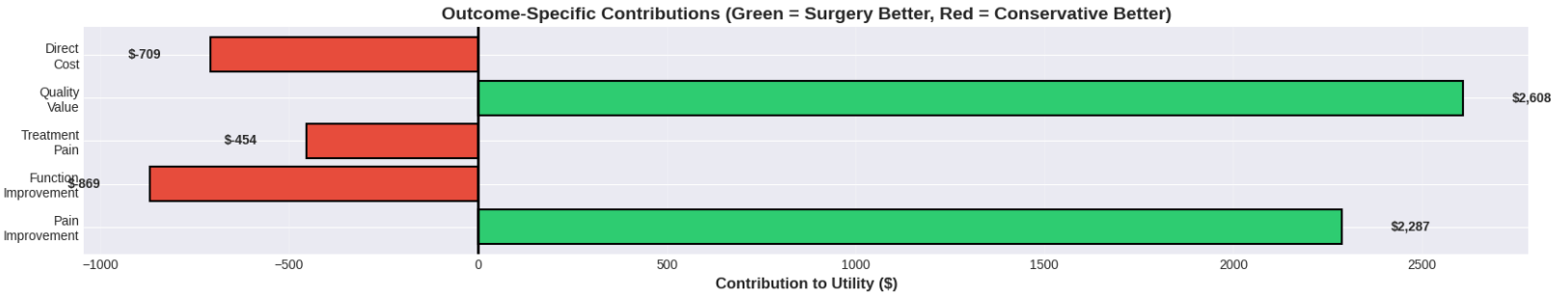
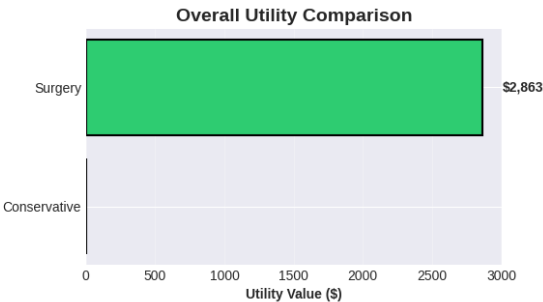
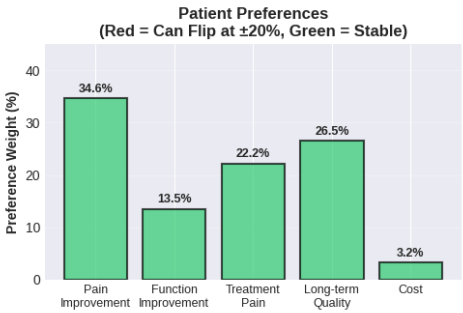


# Prototype (1/2)

Recommendation

**RECOMMENDED:**  
Surgery

**Confidence: Moderate**



SENSITIVITY SUMMARY

Level: Very Low (Robust)  
Flippable: 0/5

Base: Surgery  
Utility: \$2,862.99

STABLE  
All preferences stable at ±20% variation

Verify critical preferences carefully with patient!

KEY CONSIDERATIONS FOR SHARED DECISION MAKING:

Trade-offs:

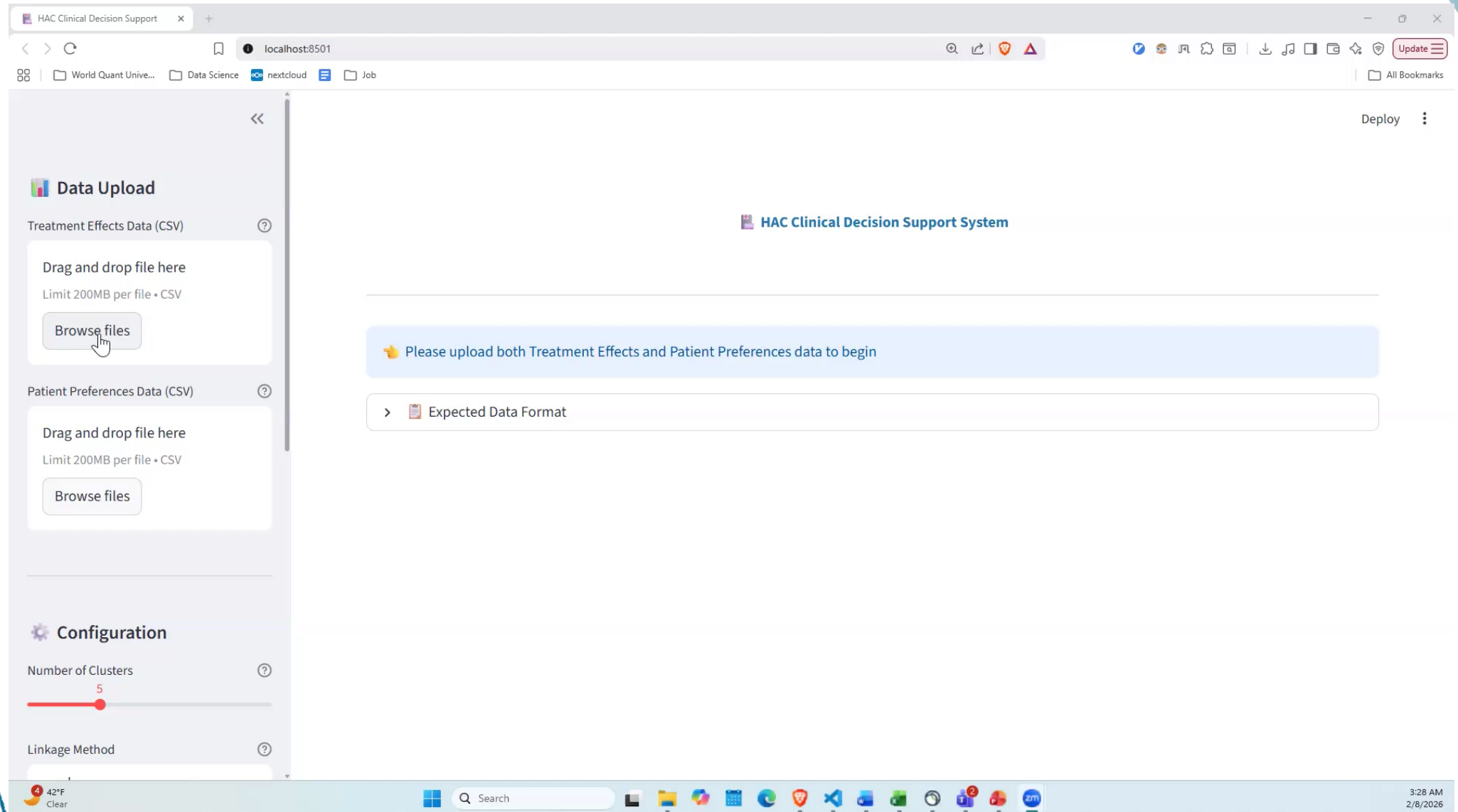
1. Surgery shows \$6,605 advantage in pain improvement value
2. Conservative care shows \$6,453 advantage in function improvement value
3. Surgery shows \$9,852 advantage in additional quality value
4. Conservative care shows \$2,044 advantage in treatment pain cost
5. Conservative care shows \$21,995 advantage in direct cost

Clinical Qualifications:

1. Mixed outcomes: Surgery is better in some areas but worse in others. Discuss specific trade-offs with patient.



# Prototype (2/2)



# Clinical Usage (1/2)

## Explanation to Doctor

### 1. Start with predicted patient outcomes

For each patient, we estimate how **surgery vs conservative care** would change: long-term pain, short-term pain, function, rehab time, and out-of-pocket cost.

### 2. Put everything on the same scale

Because these outcomes use different units, we convert them into a **common dollar-value scale** (so we can compare “pain relief” vs “time” vs “cost” fairly).

### 3. Apply what the patient values most

Patients assign priority points to the five outcomes. The tool uses those priorities to weight the outcomes.

### 4. Compare the two options

The CDS sums the weighted values into a single score:

- If the score favors surgery → recommend **Surgery**
- Otherwise → recommend **Conservative care**

### 5. Show confidence and trade-offs

The CDS flags **borderline cases** (small difference) and lists which outcomes drive the result, so you can discuss trade-offs with the patient.

### 6. Check stability

We test whether the recommendation changes if patient preferences shift slightly—if it flips easily, we label it **sensitive**, meaning the decision depends strongly on preferences.

# Clinical Usage (2/2)

## Note for Doctor

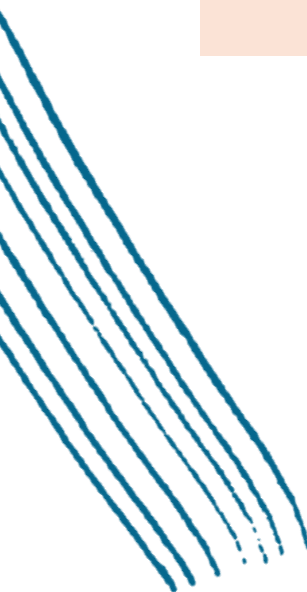
1. **When to use:** At the treatment discussion visit (after diagnosis and baseline assessment) to compare surgery vs conservative care for that patient.
2. **What it provides:** A patient-specific summary of expected differences across pain, function, rehab time, and cost—plus a recommendation and confidence level.
3. **How to use in the visit:** Review the trade-offs (what improves vs what worsens). If borderline/low confidence, treat it as a shared decision case—ask the patient what matters most and confirm preferences.
4. **Use the output to document:** “Recommendation aligns with patient priorities” or “Patient chose differently after discussion.”
5. **Safety check:** If the recommendation is sensitive (flips with small preference changes), emphasize that the decision depends heavily on values and revisit preference weights.



# Conclusion



*By combining predicted effects with patient preferences on a common value scale, **our CDS provides transparent, patient-centered recommendations to support—not replace—clinical judgment.***



# Thank you

